Lab Problem 4

Objective:

To build and evaluate logistic regression models for predicting the probability of coronary heart disease (CHD) using the **SAheart** dataset, identify statistically significant risk factors, and assess model performance using classification metrics.

Dataset Used:

- The ***SAheart*** dataset consists of medical records of male patients from a high-risk coronary heart disease (CHD) region in the Western Cape, South Africa, with approximately two control cases for each CHD case.
- Some clinical measurements were collected after medical interventions, such as blood pressure reduction programs.
- The dataset is a tab-separated CSV file containing clinical, behavioral, and demographic attributes (**e.g., blood pressure, cholesterol, tobacco use, obesity, alcohol consumption, age, and family history)** with **chd** as a binary response variable indicating CHD status.

Task:

You must perform the following tasks in order:

Task 1

- Load the SAheart dataset and build a **logistic regression model** to predict the probability of coronary heart disease,

  $$P(chd = 1),$$

  using all predictor variables except the response variable **chd.**

Task 2

- Analyze the logistic regression model from Task 1 to identify the **statistically significant features. Construct** a **new logistic regression model** using only these significant predictors.

Task 3

- From the coefficients of the logistic regression model obtained in Task 2, determine:
  - ✓ Which predictors **increase** the probability of coronary heart disease.
  - ✓ Which predictors **decrease** the probability of coronary heart disease.

Task 4

- Determine an appropriate **cut-off probability** for classifying coronary heart disease cases using the logistic regression model and justify the choice.

Task 5

- Using the cut-off probability obtained in Task 4:
  - ✓ Construct the **confusion matrix** for the model.
  - ✓ Compute and report the **precision** and **recall** for CHD cases (chd = 1).